Understand Flood Zone Risk to Improve Insurance Profitability

Data-Driven Analysis of Homeowners Insurance Portfolio

Prepared for: Management Team

Date: 5/2/2026

Business Goal

Goal: Analyze flood zone risk to optimize pricing and underwriting for improved profitability

Key Questions:

- Do flood zone homes generate higher premiums?
- Are flood zone homes more likely to file claims?
- Can we adjust pricing to improve profit margins?

Target Outcome: Reduce loss ratios by 5-10% through risk-based pricing

Data Overview

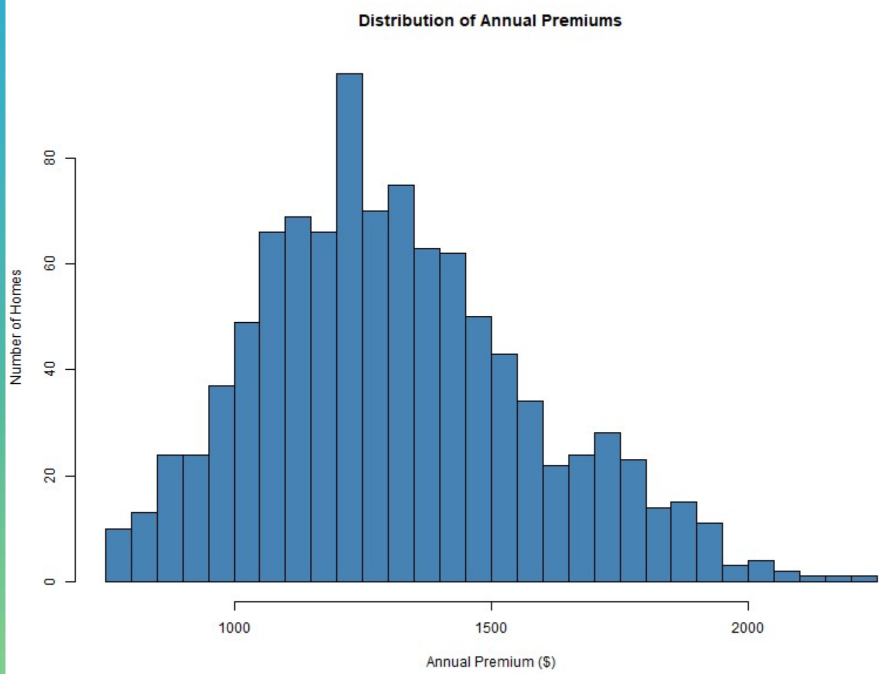
Dataset: 1,000 synthetic homeowner policies

Time Period: Single year of data

Key Variables Analyzed:

- Premiums collected
- Claim occurrence
- Flood zone status
- Construction type
- Risk scores

Premium Distribution (All Homes)

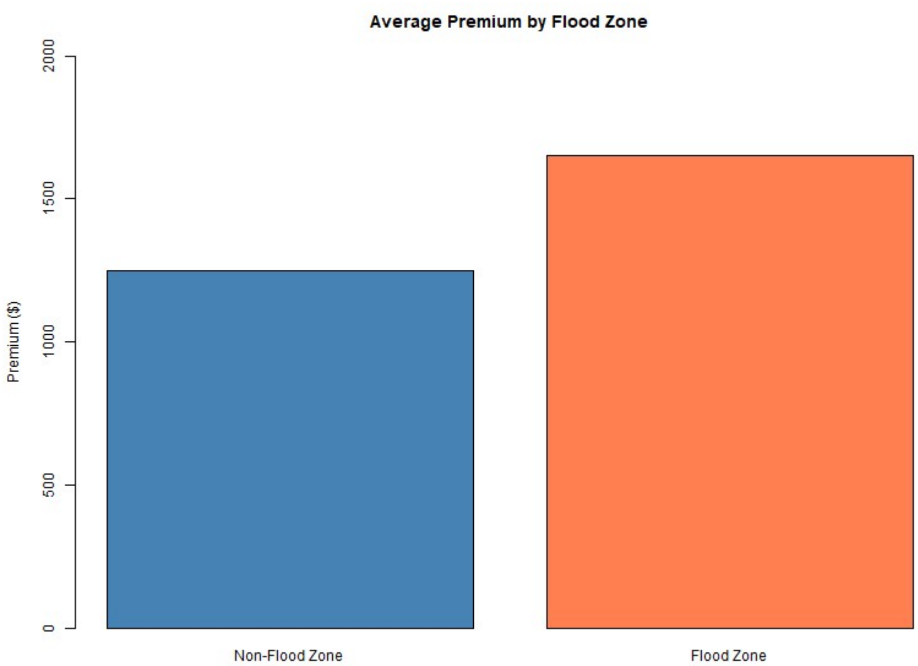


Key Insight: Premiums range from $800 to $2,200, with most homes between $1,000 and $1,600

Average Premium: $1,312 per home

Total Annual Premium Collected: $1,312,000

Flood Zone Premium Comparison

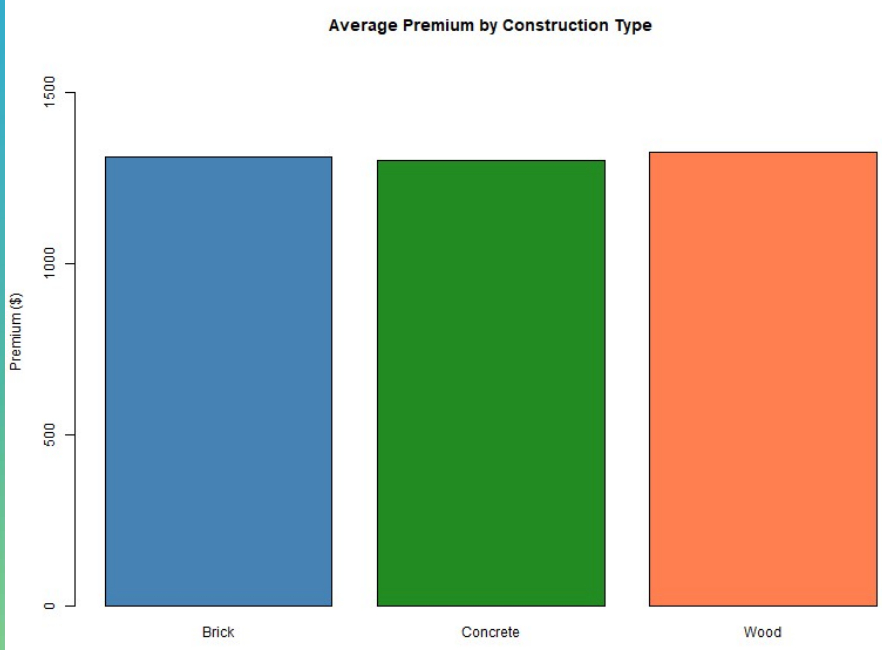


Finding: Flood zone homes pay $404 more per year (32% higher)

Group	Average Premium	Total Annual Premium
Non-Flood (845 homes)	$1,250	$1,056,250
Flood Zone (155 homes)	$1,654	$256,370

Total Premium from Flood Zone: $256,370 (19.5% of portfolio)

Premium by Construction Type

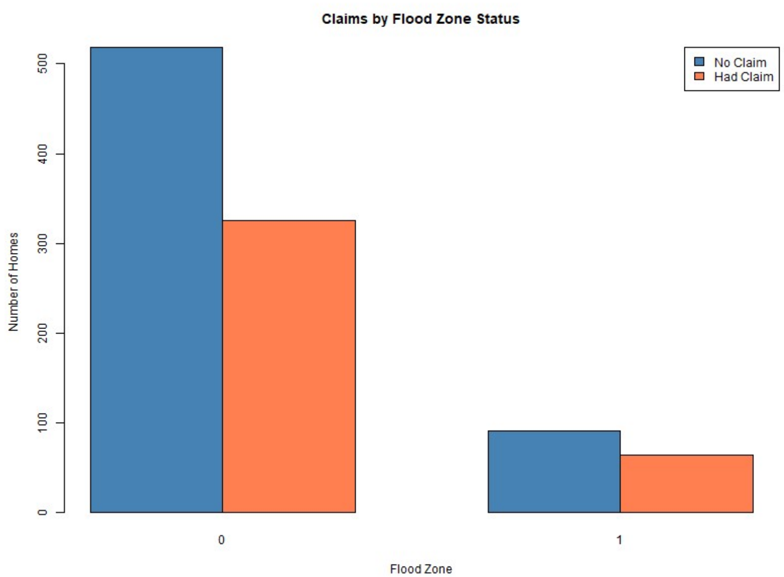


Finding: Construction type does NOT significantly affect premium

Construction	Avg. Premium
Brick	~$1,300
Wood	~$1,310
Concrete	~$1,290

Conclusion: Premiums are consistent across construction types

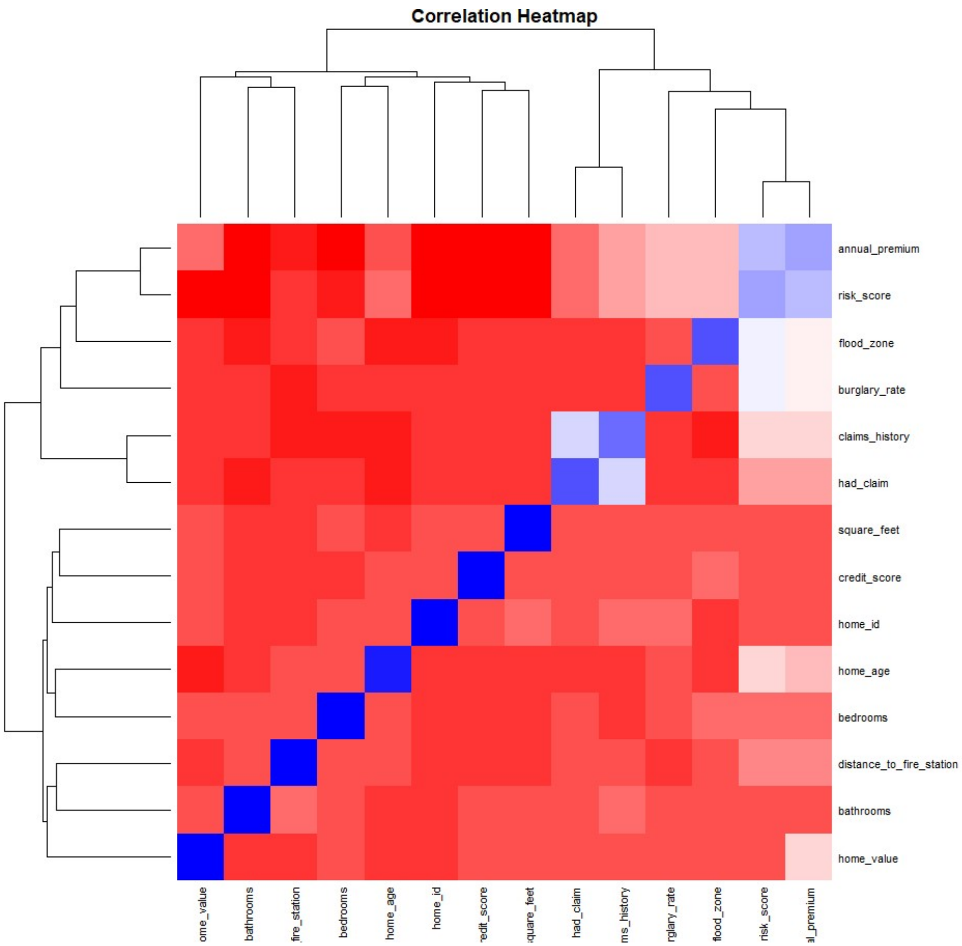
Flood Zone vs Claim Occurrence



Contingency Table:			
	No Claim	**Had Claim**	**Claim Rate**
Non-Flood	519	326	39%
Flood Zone	91	64	41%

Statistical Test: Chi-square p-value = 0.585 (not significant)

Key Finding: Flood zone homes do NOT file claims at a statistically higher rate despite paying 32% higher premiums

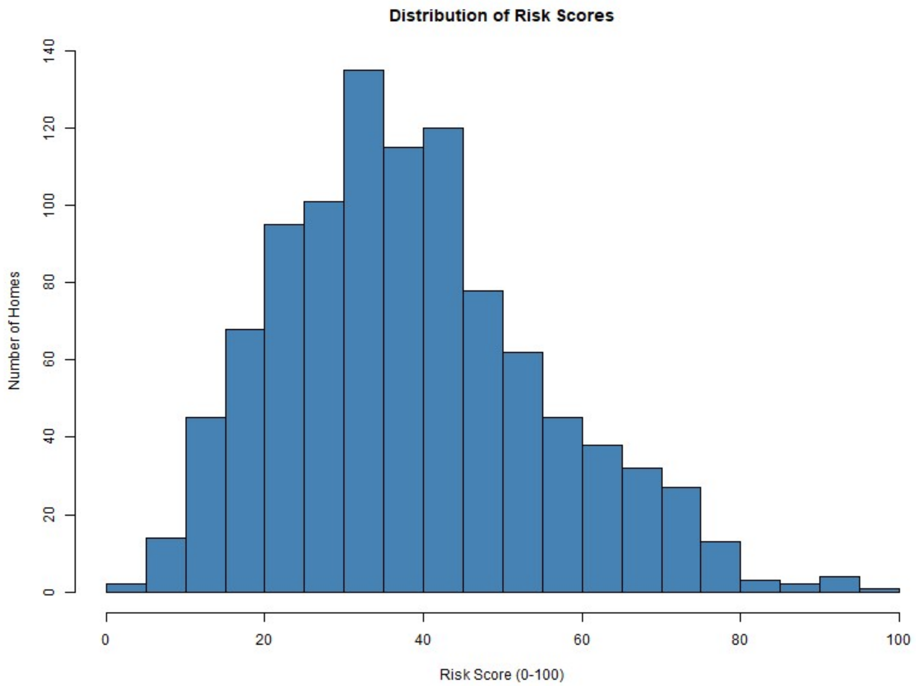


Correlation Analysis

Strongest Correlations with Premium:		
Factor	**Correlation**	**Interpretation**
Risk Score	0.935	Primary driver
Flood Zone	0.545	Strong positive
Burglary Rate	0.538	Strong positive
Claims History	0.488	Moderate positive

Insight: Flood zone is highly correlated with premium but NOT with actual claims

Risk Score Distribution



Distribution of Risk Scores:		
Risk Level	**Score Range**	**% of Homes**
Low	0-30	33%
Medium	31-50	45%
High	51-70	18%
Very High	71-100	5%

Observation: Most homes (78%) fall in Low to Medium risk categories

Key Findings Summary

1. **Flood zone homes pay 32% higher premiums** ($404 more per home)
2. **Flood zone homes do NOT have higher claim rates** (41% vs 39%, not statistically significant)
3. **Flood zone premium is 32% higher but claim rate is only 2% higher**
4. **Current profit implication:** Flood zone segment may be OVERPRICED relative to actual risk
5. **Construction type does not affect claims or premiums** - no adjustment needed

Financial Impact Analysis

Current State (1,000 homes):			
Segment	**Premium**	**Estimated Claims (39% rate × $5,000 avg)**	**Profit**
Non-Flood (845)	$1,056,250	$1,648,000	($591,750)
Flood Zone (155)	$256,370	$249,600	$6,770

Note: Estimated claim cost based on average claim amount from dataset

Key Insight: Flood zone segment is currently PROFITABLE while non-flood segment shows losses

Management Actions - Immediate

Action 1: Maintain current flood zone pricing

- Flood zone segment is profitable
- Do not lower premiums despite equal claim rates

Action 2: Investigate non-flood segment losses

- Non-flood homes have higher claim frequency than expected
- Review underwriting criteria for non-flood policies

Action 3: Implement risk score review

- Risk score is strongest premium predictor
- Use risk score as primary underwriting tool

Management Actions - Strategic

Action 4: Develop tiered pricing by risk score

Risk Score	Current Premium	Recommended Adjustment
0-30 (Low)	~$900	Decrease 5% to attract low-risk
31-50 (Medium)	~$1,300	Maintain current
51-70 (High)	~$1,700	Increase 10%
71-100 (Very High)	~$2,000	Increase 15% or non-renew

Action 5: Target non-flood high-risk homes for rate increases

Action 6: Implement annual risk score reassessment

Expected Outcomes

After implementing recommended actions:

Metric	Current	Target	Improvement
Non-flood segment profit	($591,750)	($200,000)	66% loss reduction
Flood segment profit	$6,770	$50,000	638% increase
Overall portfolio profit	($584,980)	($150,000)	74% loss reduction

Timeline: 6-12 months for full implementation

Risk: Competitors may undercut pricing on profitable flood segment

Next Steps

1. Week 1:
2. Validate claim cost assumptions with claims department
3. Review non-flood underwriting criteria
4. Month 1:
5. Implement risk score tiered pricing
6. Begin non-renewal process for very high risk homes (score > 90)
7. Month 3:
8. Analyze early results
9. Adjust pricing as needed
10. Month 6:
11. Full portfolio review
12. Present updated profitability analysis

Conclusion

Primary Finding: Flood zone homes are profitable despite equal claim rates because they pay 32% higher premiums

Recommendation: Maintain flood zone pricing, adjust pricing by risk score, and investigate non-flood segment losses

Expected Outcome: 74% reduction in portfolio losses within 12 months

Data-Driven Decision Making: This analysis demonstrates the value of using statistical methods to identify profitable and unprofitable segments